MILITARY SPECIFICATION

FILM, POLYESTER

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers three types of polyester film for use as burst diaphrams.

1.2 Classification. Polyester film shall be of the following types:

Type I	0.002 inch thick
Type II	0.003 inch thick
Type III	0.005 inch thick

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization/Documentation Division (524), Indian Head, Maryland 20640, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the acquiring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DOD adopted shall be the issue listed in the current DODISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D374-79	Thickness of Solid Electrical Insulation, Test Methods for
ASTM D882-80A	Tensile Properties of Thin Plastic Sheeting Test Methods for

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL MOTOR FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, NW, Washington, DC 20036).

UNIFORM FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606).

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Material. The material shall be a polyethylene terephthalate film.

3.2 Thickness. The thickness of the material shall conform to the requirements shown in Table I.

TABLE I. Thickness.

	Type I	Type II	Type III
Thickness (inches)	.002	.003	.005
Permissable variation (inches (plus or minus)	.0005	.0005	.0010

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3.3 Physical Properties. The physical properties of the material shall conform to the requirements shown in Table II.

TABLE II. Physical properties.

Property	Acceptable Values					
	Type I		Type II		Type III	
	Min	Max	Min	Max	Min	Max
Modulus of elasticity (psi)	287,000	-	298,000	-	334,000	-
Tensile strength (psi)	17,000	-	17,000	-	17,000	-
Elongation (percent)	92	-	79	-	83	-

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserv the right to perform any of the inspections set forth in the specification whe such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot. A lot shall consist of material of one type, manufactured in one uninterrupted production run, without change in process, materials, or equipment and offered for delivery in one shipment.

4.3 Sampling. One sample shall be taken from each roll or package of sheets.

4.3.1 Sample size. The sample size shall be approximately 360 square inches.

4.4 Quality conformance inspection. The material shall be subjected to all the following inspections and tests for acceptance. When specified in the contract (see 6.2), the contractor shall prepare a report giving the results obtained for all inspections and tests performed and a certified statement that the lot meets all the requirements of this specification.

4.4.1 Thickness. The thickness of the material shall be determined in accordance with method A or C of ASTM D 374.

4.4.2 Modulus of elasticity. The modulus of elasticity of the material shall be determined in accordance with method A of ASTM D 882, using a rate of grip separation of 2.0 inches per minute.

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4.4.3 Tensile strength. The tensile strength of the material shall be determined in accordance with method A of ASTM D 882, using a rate of grip separation of 2.0 inches per minute.

4.4.4 Elongation. The elongation of the material shall be determined in accordance with A of ASTM D 882, using a rate of grip separation of 2.0 inches per minute.

4.5 Marking and packaging inspection. The inspector shall ascertain that the marking and packaging conform to the requirements of Section 5 by a visual examination.

5. PACKAGING

5.1 Packaging.

5.1.1 Commercial. Unless otherwise specified, the material shall be packaged in accordance with the manufacturer's best accepted commercial practice.

5.2 Packing.

5.2.1 Commercial. Unless otherwise specified, the material shall be packed in shipping containers in such a manner as to insure safe transportation by common carrier. Containers used shall comply with National Motor Freight Classification Rules and the Uniform Freight Classification Rules or regulations of other carriers applicable to the mode of transportation.

5.3 Marking. Marking of packages shall be in accordance with MIL-STD-129 and shall include but not be limited to the following:

a. Title, number and revision letter of this specification

b. Name of manufacturer or supplier

c. Type of material

d. Lot number and date of manufacture

e. Quantity in package

f. Contract or purchase order number

6. NOTES

6.1 Intended use. The polyester film purchased in accordance with this specification is intended for use as a burst diaphragm in the Mk 28 Mod 0 Army Firing Device used in the Mk 272 Mod 0 Igniter.

6.2 Ordering data. Acquisition documents should specify the following:

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6.2.1 Acquisition requirements.

a. Title, number, revision letter and date of this specification

b. Type of material (see 1.2)

c. Quantity of material and form - rolls or sheets

d. Lot or batch number

e. Place and date of delivery

f. Whether a certified analysis and/or test report is required (see 4.4)

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data Requirement	Applicable DID
4.4	Test Report	DI-T-2072
4.4	Certification Data/ Report	UDI-A-23264

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DOD 5000.19L, Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

Preparing Activity:
Navy-OS

Project Number: 9330-N047

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
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	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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